JAY THAKRE

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SUMMARY

B.Tech AI & ML student who builds full-stack web platforms, RAG-based AI tutors, applied ML models and Android apps. Co-author of an IEEE Xplore paper. Seeking a software engineering or AI/ML internship.

EDUCATION

B.Tech, Artificial Intelligence & Machine Learning Expected 2028

Faculty of Engineering and Technology, AIHER (Deemed to be University), Sawangi, Wardha | CGPA: 7.96 / 10

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, Kotlin, Dart, HTML/CSS

Frameworks: React, React Native, Flutter, Node.js/Express, FastAPI, Flask, Tailwind CSS, Android (Jetpack Compose)

Data & APIs: SQL, MongoDB, Redis, Firestore, REST APIs, OpenAPI, Firebase, Gemini API, AWS S3, Razorpay

AI / ML: Machine learning, CNNs, TensorFlow/Keras, RAG (embeddings, ChromaDB, LLM tutors), XGBoost, structured LLM output

Core: Data Structures & Algorithms, full-stack development, Git/GitHub, Docker, JWT/RBAC API security, unit testing (Vitest), IoT prototyping

PROJECTS

CareTrack: Clinic Follow-up & Patient Reminder App | *Flutter, Dart, Firebase, Cloud Functions, Gemini API*

- Built a Flutter Android app (admin, doctor roles) with real-time Firestore sync, a follow-up dashboard (due today, 3 days, 7 days, overdue) and per-doctor data isolation using Firestore security rules and Cloud Functions.
- Designed a Gemini API flow that extracts medicines, dosages and durations from prescription photos and PDFs with mandatory doctor verification, and built email, WhatsApp and push reminders with idempotent scheduling. (Code: github.com/jayy1506/caretrack)

MedPulse: Clinical Nursing LMS & RAG AI Tutor | *React 19, Node.js, Express, MongoDB, Gemini, Redis, Docker*

- Built a clinical nursing LMS with a RAG AI tutor (Gemini text-embedding-004 retrieval, Gemini 1.5 Flash answers).
- Backed it with MongoDB, Redis and Firebase Admin; added Pino logging, Swagger (OpenAPI 3.0) docs and Docker. (Live: medpulse-nursing.vercel.app, Code: github.com/jayy1506/Nursing)

Adaptive-Retrieval RAG Router | *Python, FastAPI, XGBoost, ChromaDB, Sentence-Transformers, Docker*

- Built an XGBoost query router (14 linguistic features, MiniLM embeddings) that picks the retrieval chunk budget ($k \in \{0, 3, 8, 15\}$), reaching 95.3% accuracy at under 3 ms latency.
- Added a 2-stage confidence fallback ($\tau = 0.65$) and sufficiency escalation over 10,900+ ChromaDB passages, cutting token cost with 0.0% under-retrieval; served via FastAPI with SQLite SHA-256 LLM caching, a web dashboard and Docker. (Live: adaptive-rag-router.onrender.com, Code: github.com/jayy1506/Adaptive-Rag-Router)

AyurLab: AI-Powered Ayurveda Virtual Lab & Learning Platform | *React 19, Node.js, Express 5, MongoDB, Gemini, AWS S3, Docker*

- Built an interactive compounding simulator (React 19, @dnd-kit) with a deterministic grading engine, and a server-side RAG Gemini tutor grounded in classical Ayurvedic texts, with a rotating key pool and emergency-triage guardrails.
- Added in-browser exam proctoring (MediaRecorder, MediaPipe) with chunked uploads to AWS S3; secured the API with Zod, Helmet, Redis rate limiting and JWT/refresh-cookie auth; 30-test Vitest suite. (Live: ayurlab-1.onrender.com, Code: github.com/jayy1506/Ayurlab)

SafarX: Corridor Shared-Seat Mobility Platform | *TypeScript, Node.js, Express, MongoDB, Razorpay, Docker*

- Built a shared-transit platform (Passenger, Driver, Admin portals) with a segment-based engine that prevents multi-hop double-booking.
- Added atomic MongoDB seat holds with a 10-minute reaper job, Razorpay with HMAC-SHA256 verification and idempotent webhooks, and a 4-digit boarding OTP. (Live: safarx-ikmw.onrender.com, Code: github.com/jayy1506/SafarX)

Reveille: AI Voice Alarm & Daily Mission Tracker | *Kotlin, Jetpack Compose, Python, FastAPI, Gemini API*

- Built a native Android alarm app (Kotlin, Jetpack Compose, Material 3) that wakes the screen over the lockscreen via AlarmManager.setAlarmClock, with SpeechRecognizer and TextToSpeech for a hands-free voice debrief.
- Built a FastAPI backend on Gemini 2.5 Flash with Pydantic JSON schemas, multi-key failover and 8-second timeouts, plus a regex-based offline fallback for task parsing. (Code: github.com/jayy1506/Reveille)

CapTrack: Smart Inhaler Cap | *Prototype (PBL) | ESP32/Arduino, DS3231 RTC, EEPROM*

- Designed the circuit and presentation for a low-cost, offline inhaler cap that logs dose events (RTC) and sends reminder alerts.

EXPERIENCE

Hospital Internship, Pediatrics Dept., AVBRH Hospital, Wardha Apr 2025 – May 2025

- Joined software sessions and ward visits to observe clinical workflows and identify problem statements.

Intel® Unnati Industrial Training 2025 (Intel / EdGate / AICTE) Oct 2025 – Dec 2025

- Worked on an AI-Powered Form Filling Assistant for Indian Citizen Services (guide: Dr Utkarsha Pacharaney).

LEADERSHIP & ACTIVITIES

- General Secretary (college-wide events); Chairperson, Intel Club; Treasurer, IEEE Student Chapter; Head, Pooja Committee; Cultural Co-Head, Ignitra; 3 hackathons; Destination Dr. Reddy's 3.0 (2026); NISM/SEBI Financial Literacy Quiz 2026.

CERTIFICATIONS

- **Hedera:** Certified Foundation, Business Foundation, Developer Associate (2026) | **Coursera:** IBM, Imperial, Edinburgh, Johns Hopkins courses (2025) | **Other:** Cisco Python Essentials 1 & 2; AI Tools & ChatGPT (be10x)